

RAG Methodology

Project Health Reporting Agent — Phase 1 Framework

Approach

Project health is derived from **independently computed, auditable signals** — never a single opaque score — combined with a **“worst signal wins”** rule for the hard, verifiable metrics. Every signal below was tested directly against the two sample project files before being adopted; none is assumed from column names or vendor labels. Where a signal is missing, contradictory, or unreliable, the agent **says so explicitly** rather than defaulting it to a safe-looking value.

All task-level aggregate metrics (overdue counts, percentages) are computed at **leaf-task granularity only** — tasks with no children in the project hierarchy — because phase/summary rows frequently duplicate their single child task's name, dates, and status (confirmed: 8 duplicate rows in one sample file, 142 in the other); counting both would double-count the same real-world work.

Signals and RAG Mapping

Signal	What it measures	Green	Amber	Red
Schedule Slippage	% of active (Not Started / In Progress) leaf tasks past their End Date and not Completed	< 10%	10–20%	> 20%
Critical-Path Override	Overdue active tasks with ~zero Total Float (i.e., on the critical path)	0	1 → forces ≥ Amber	2+ → forces Red
Milestone Health	% of major phases (top-level hierarchy nodes, labeled by name where the source data provides one) overdue and not Completed	0%	1–15%	> 15%
Data Integrity Check	Any task whose free-text comment contradicts its own recorded Status / % Complete — checked on every commented task, active or not	none found	—	any hit forces ≥ Amber, reported by name
Blockers	Overdue-task lens, supplemented by blocker language in stakeholder comments	—	Flagged as a lagging indicator — narrative caveat, not a color driver	
Stakeholder Sentiment	Tone/urgency of free-text comments, scored where any comment data exists	Reported only where data exists; never defaulted to neutral		

Signal	What it measures	Green	Amber	Red
Budget Burn	—	Out of scope — no cost/budget data exists in either source file		

Overall project status = the **worst** of {Schedule Slippage, Milestone Health, Critical-Path Override, Data Integrity Check}. Sentiment and Blocker findings are always reported alongside the color as narrative context — testing showed they are weaker, laggy predictors of real risk than the schedule-based signals, so they inform the *reasoning*, not the *color*.

Conflicts are surfaced, never averaged away. If the source file's own fields (Schedule Health, At Risk, RAG where present) disagree with the agent's computed status, that disagreement is reported explicitly. Averaging a Green and a Red into a Yellow would represent nobody's actual assessment — and there is direct evidence this disagreement is common and meaningful, not noise to smooth over.

Why the Data Integrity Check Exists

Every other signal above only examines *active* tasks, by design — a Completed task can't be “overdue.” Auditing the free-text comments turned up a case that logic structurally cannot catch: two tasks recorded as **Status = Completed, % Complete = 100%** carried the comment “**Yet to receive Sign off.**” Neither the file's own RAG nor Schedule Health flagged this contradiction. Since a task marked Completed is invisible to every other signal computed here, this required a dedicated check that runs independent of task status — the one place in the audit where something was found that the original design couldn't have caught even in principle, not just something it confirmed.

Assumptions and Verified Data-Quality Findings

- **Budget data does not exist** in either sample file — no cost columns anywhere. Treated as fully out of scope rather than proxied.
- **Milestone/Phase and Area fields are almost entirely blank.** Milestone health is computed from the task hierarchy (top-level phase rollups); where the source data does supply a real phase name (confirmed present on 9 of 14 major phases in one file — e.g. “UAT,” “Go Live,” “Hypercare”), that name is used in the output for readability.
- **Priority and “At Risk?” are populated on well under 1% of rows** in both files and are not used as standalone signals — imputing a default would create fake-looking data indistinguishable from real input. Where At Risk? is present (4 rows, one file), it does directionally align with non-Green status, but the sample is too small to generalize.
- **Stakeholder comments come from two independent channels, not one** — a row-level comments column and a separate comments log, referencing entirely different tasks in the one file that has either. Both are combined into a single sentiment/blocker input; treating only one would silently drop real signal.
- **The existing Variance column is unreliable** — verified against a direct End Date vs. Baseline Finish recomputation and found to disagree on 21% of rows in one file and 73% in the other. The agent computes slippage independently rather than trusting this field.

- **The existing Schedule Health column does not derive from Critical, Total Float, or Variance** — tested directly against all three, none correlate. It tracks something close to “is this task currently overdue,” but inconsistently between the two files (96% precision / 71% recall in one, 78% / 100% in the other), so it is used only as an external comparison point, never as an input.
- **The RAG column present in one file is a retrospective delivery grade, not a live risk indicator** — 206 of 250 labeled tasks were already Completed at the time of labeling, and it correlates poorly with any forward-looking signal tested. It is not used as ground truth for validating the framework.
- **“Overdue” is a lagging indicator**, confirmed by testing: only 4 of 50 PM-flagged-Red tasks in the sample were actually overdue by this definition — most real risk gets flagged before a deadline is missed. Blockers are therefore a supporting narrative signal, not a primary driver of status.
- **Duplicate rows are common** (8 in one file, 142 in the other) where a phase header and its single child task share identical values — handled by restricting all aggregation to leaf-level tasks.
- Several columns (No.of days Until Today, Target Start/End Date, Project Category, and — in one file — a Level column that duplicates Ancestors value-for-value) are either **dead formulas returning #UNPARSEABLE** on every row or exact duplicates of another column; a handful of Predecessors entries are broken #REF links. All are dropped or ignored at ingestion rather than treated as ordinary missing data.